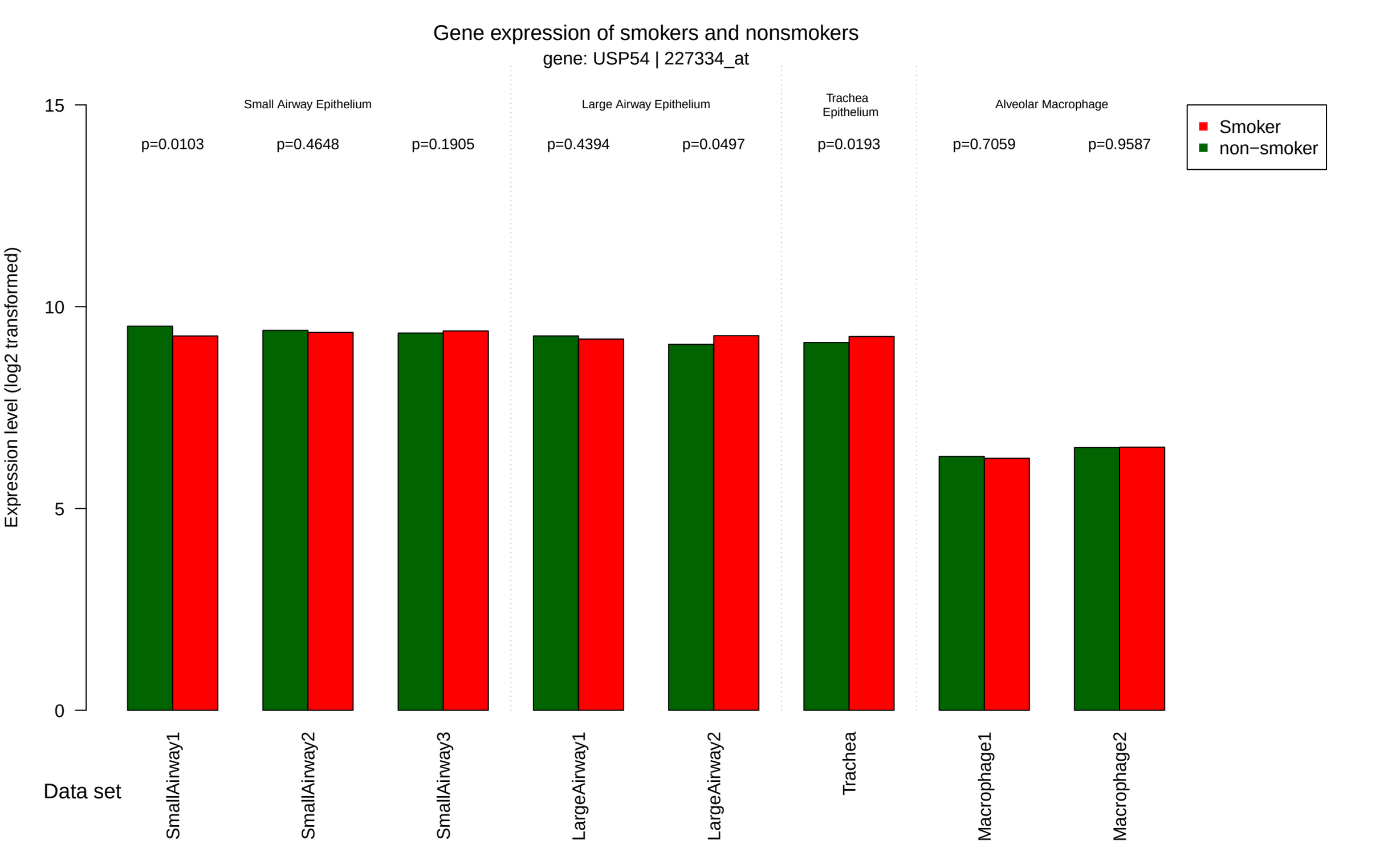
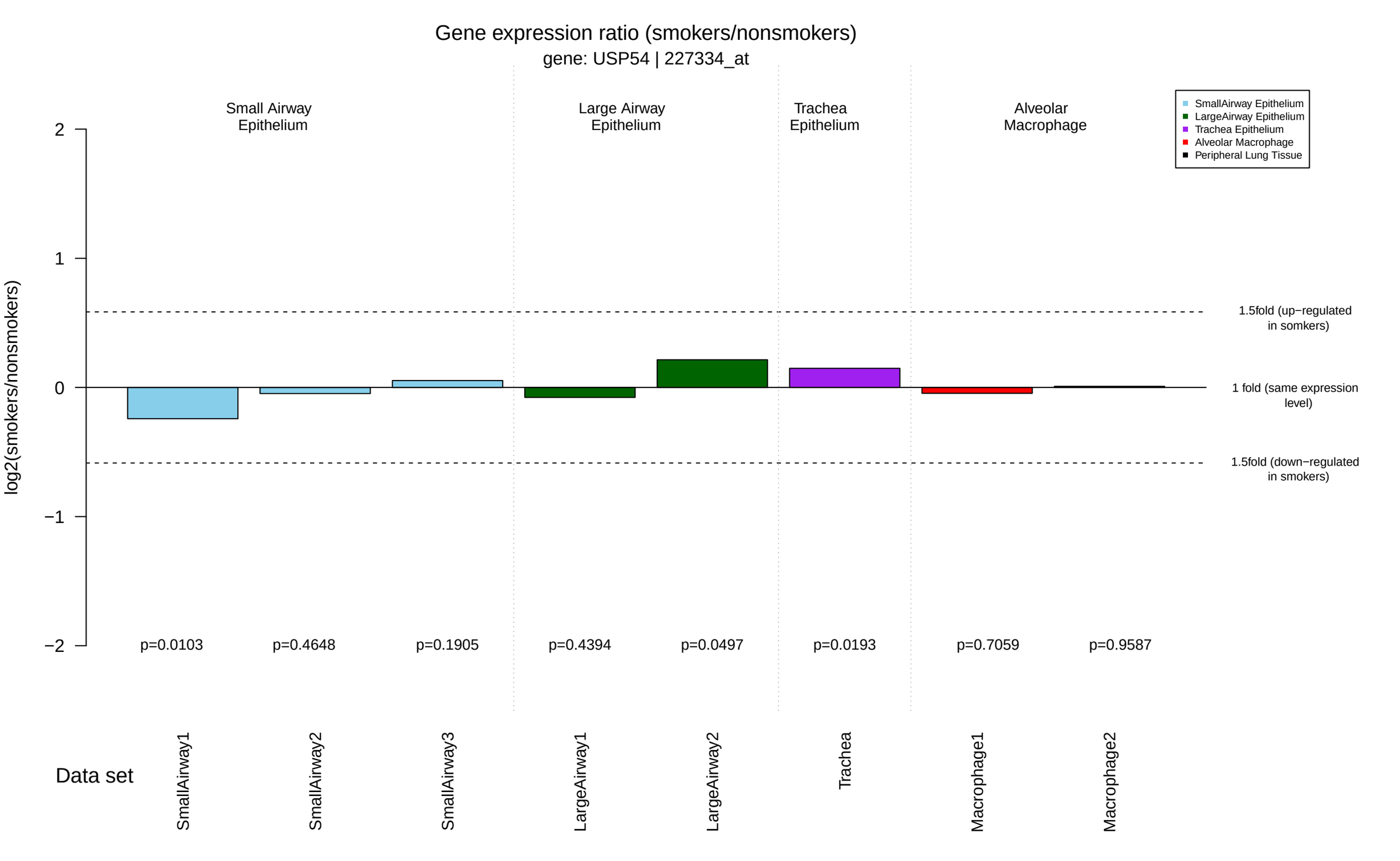
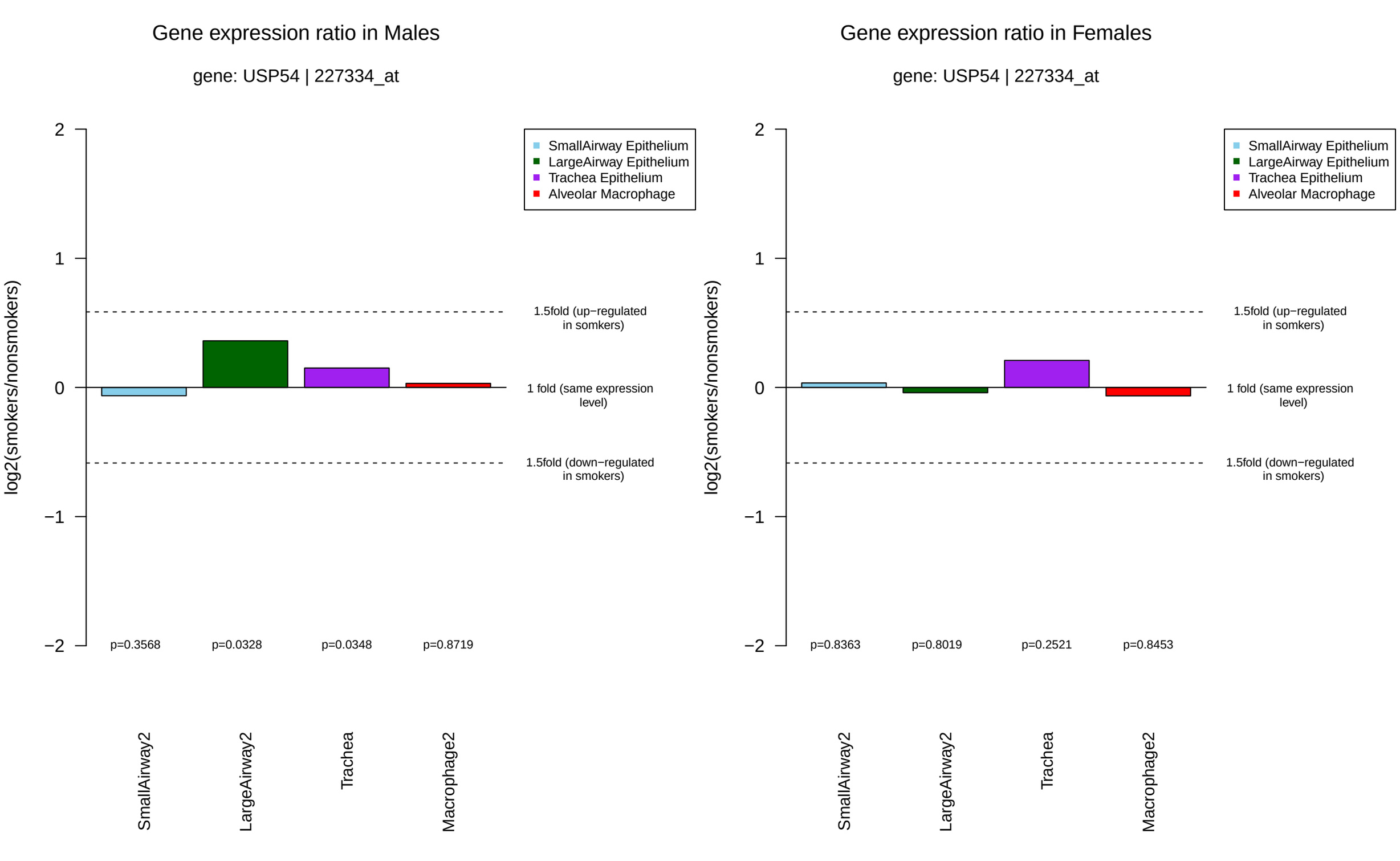
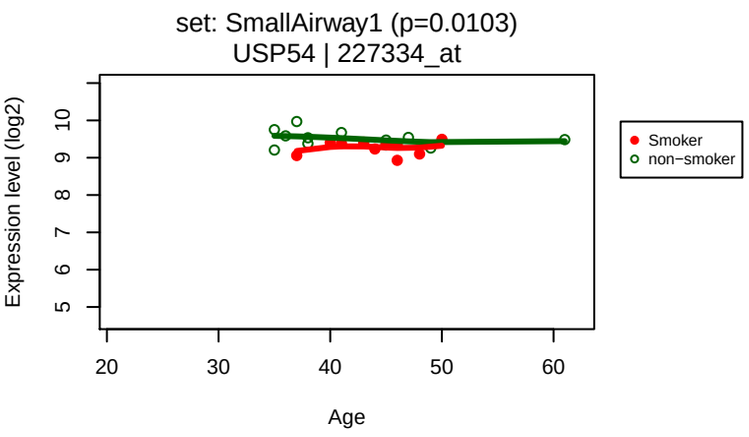
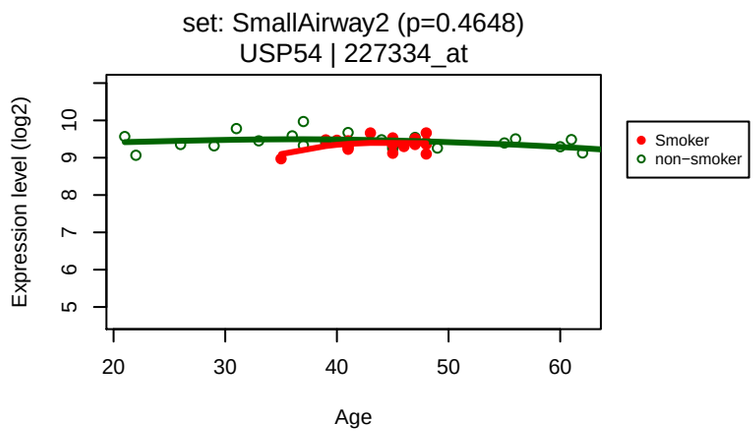
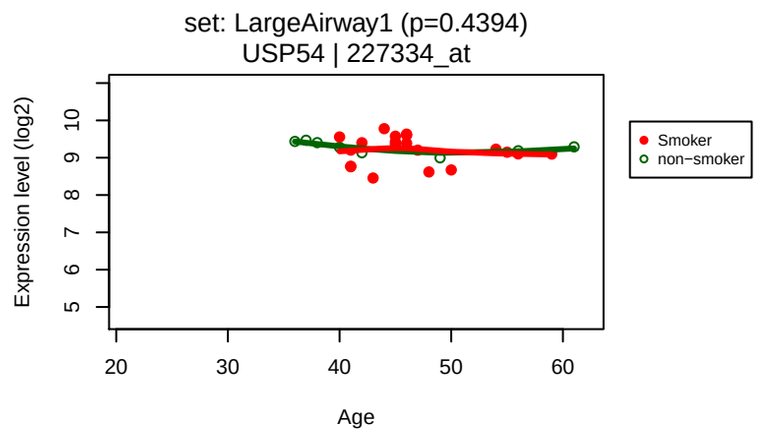

Gene expression of smokers and nonsmokers
gene: USP54 | 227334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0103
p=0.4648
p=0.1905
p=0.4394
p=0.0497
p=0.0193
p=0.7059
p=0.9587
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

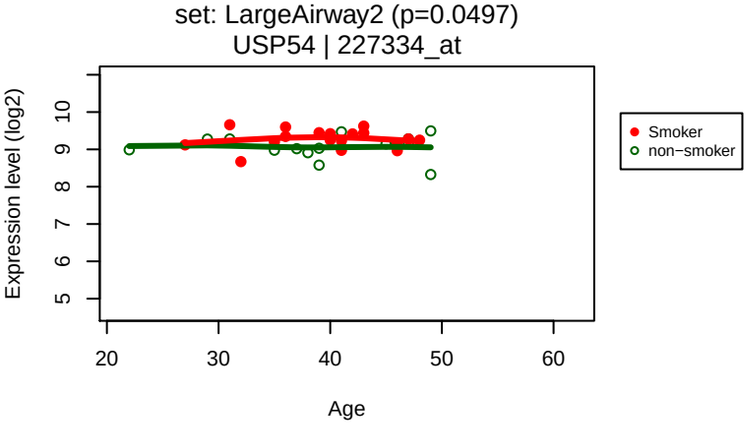

Gene expression ratio (smokers/nonsmokers)
gene: USP54 | 227334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0103
p=0.4648
p=0.1905
p=0.4394
p=0.0497
p=0.0193
p=0.7059
p=0.9587
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

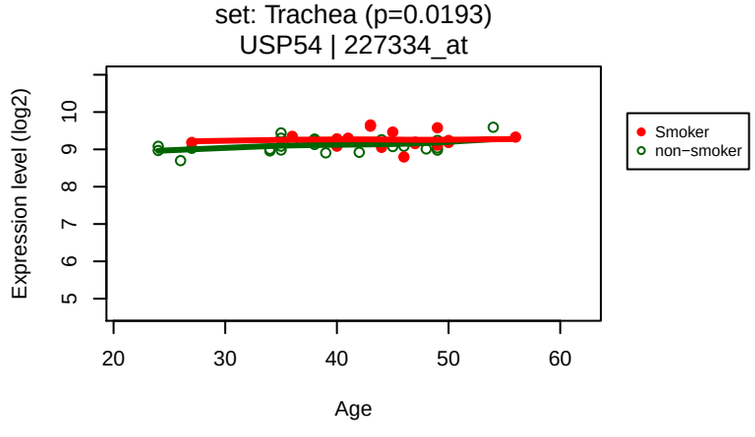

Gene expression ratio in Males
gene: USP54 | 227334_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3568
p=0.0328
p=0.0348
p=0.8719
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP54 | 227334_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8363
p=0.8019
p=0.2521
p=0.8453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

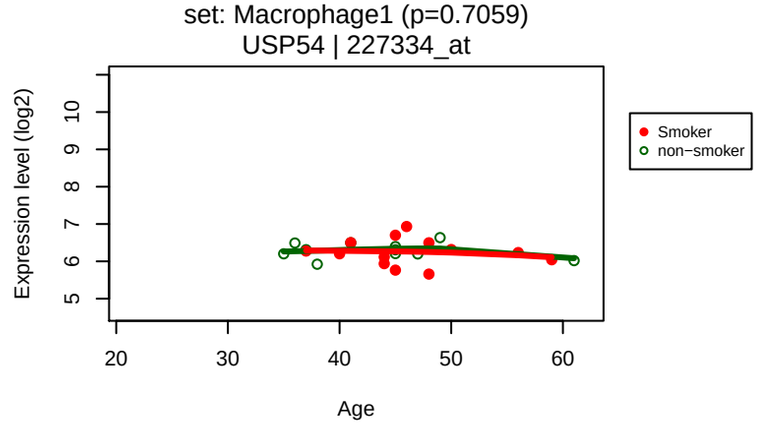

set: SmallAirway1 (p=0.0103)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker

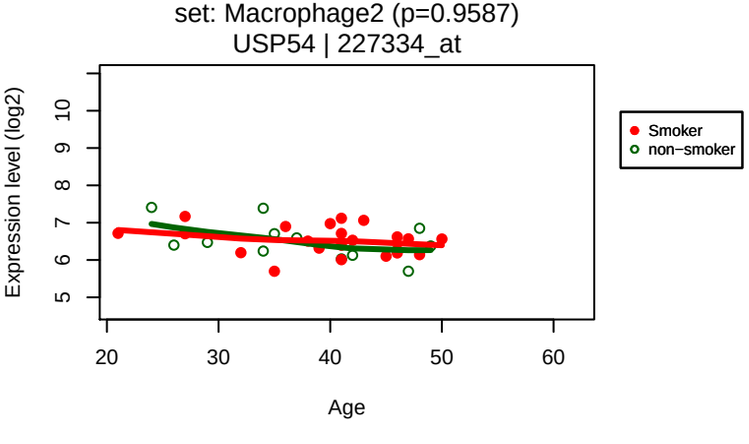

set: SmallAirway2 (p=0.4648)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker

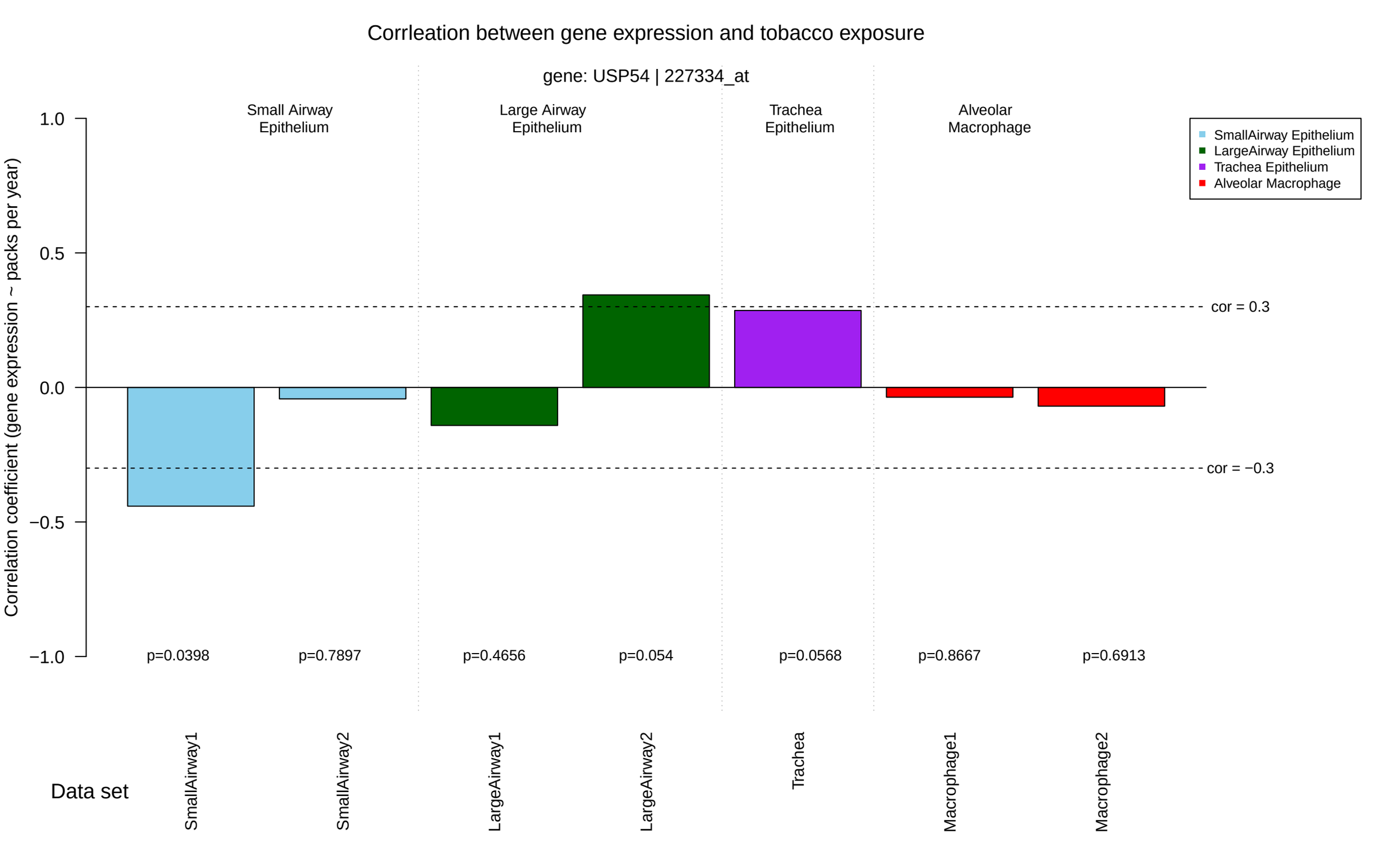

set: LargeAirway1 (p=0.4394)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0497)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0193)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7059)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9587)
USP54 | 227334_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP54 | 227334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0398
p=0.7897
p=0.4656
p=0.054
p=0.0568
p=0.8667
p=0.6913
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2